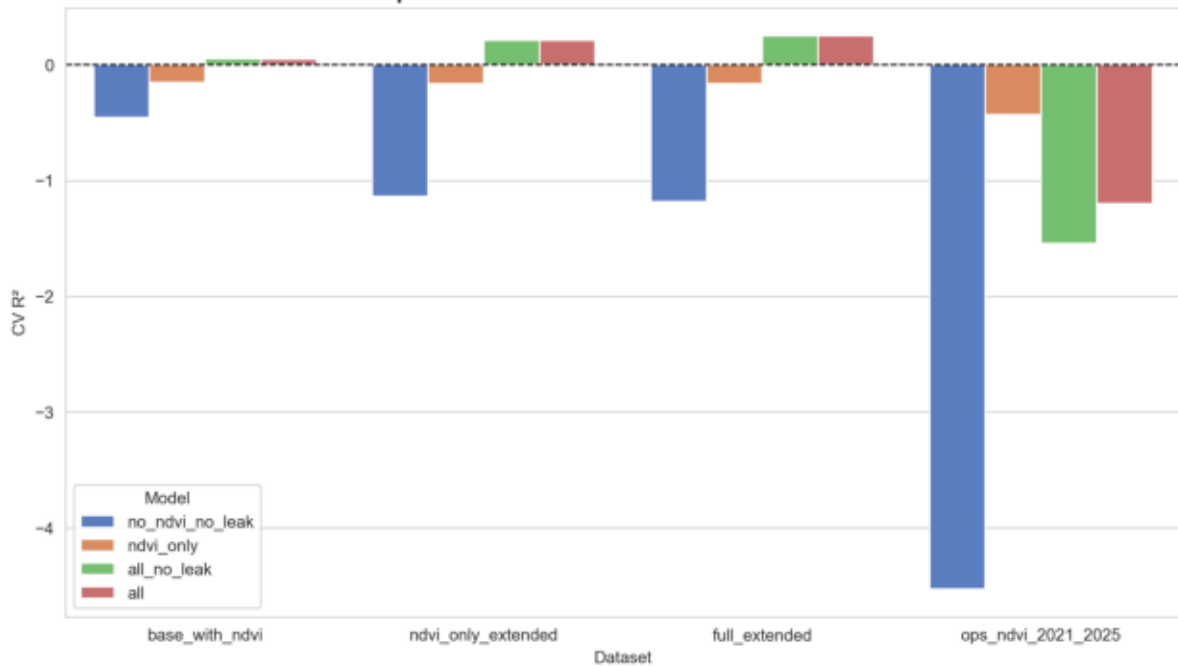


Models comparison summary

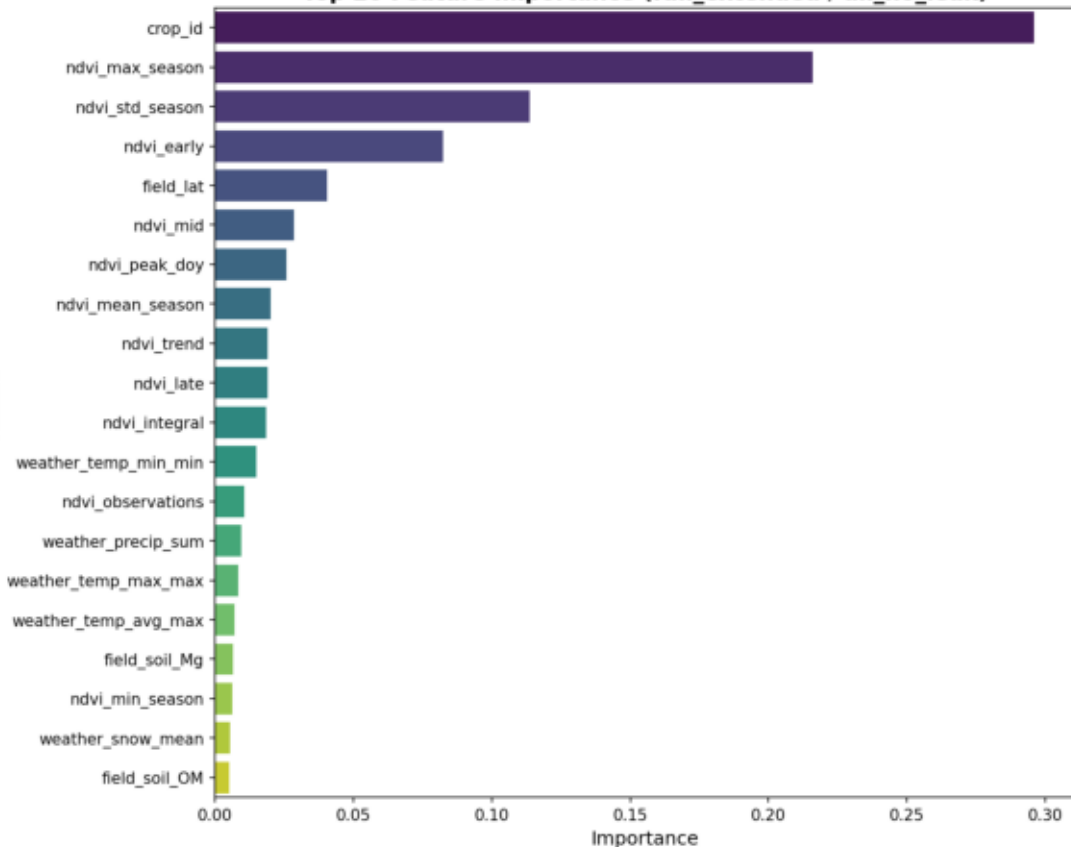
Dataset	Model	N obs	CV R2	CV RMSE (t/ha)	CV MAE (t/ha)	CV MAPE (%)	Train R2	Train RMSE (t/ha)
base_with_ndvi	no_ndvi_no_leak	440	-0.457	7.45	5.53	25.4	0.621	4.73
base_with_ndvi	ndvi_only	440	-0.153	6.99	5.26	23.6	0.988	0.85
base_with_ndvi	all	440	0.05	6.17	4.62	21.0	0.989	0.79
base_with_ndvi	all_no_leak	440	0.051	6.16	4.58	20.9	0.991	0.72
ndvi_only_extended	no_ndvi_no_leak	441	-1.138	9.09	7.1	33.5	0.963	1.49
ndvi_only_extended	ndvi_only	441	-0.163	6.97	5.21	23.5	0.985	0.94
ndvi_only_extended	all	441	0.211	5.59	4.03	17.9	0.995	0.54
ndvi_only_extended	all_no_leak	441	0.211	5.59	4.03	17.9	0.995	0.54
cps_ndvi_2021_202	no_ndvi_no_leak	136	-4.527	11.12	9.01	32.3	0.999	0.22
cps_ndvi_2021_202	ndvi_only	136	-0.43	6.09	4.61	15.9	1.0	0.1
cps_ndvi_2021_202	all	136	-1.199	7.46	5.73	20.1	1.0	0.07
cps_ndvi_2021_202	all_no_leak	136	-1.542	7.93	6.04	21.2	1.0	0.05
full_extended	no_ndvi_no_leak	441	-1.18	9.07	7.01	32.8	0.963	1.49
full_extended	ndvi_only	441	-0.163	6.97	5.21	23.5	0.985	0.94
full_extended	all	441	0.251	5.51	3.97	17.7	0.995	0.55
full_extended	all_no_leak	441	0.251	5.51	3.97	17.7	0.995	0.55

Comparison of CV R² across Datasets and Models

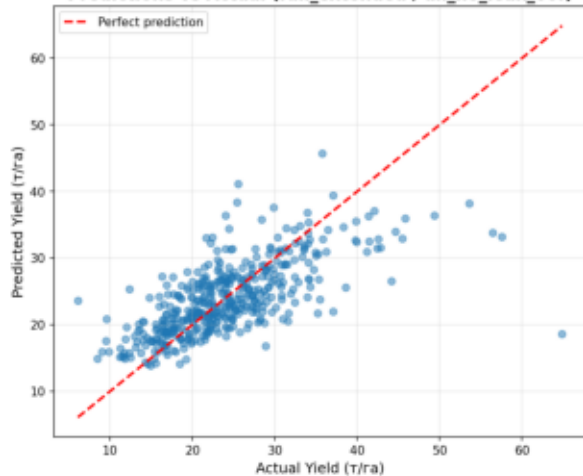


Top 20 Feature Importance (full_extended / all_no_leak)

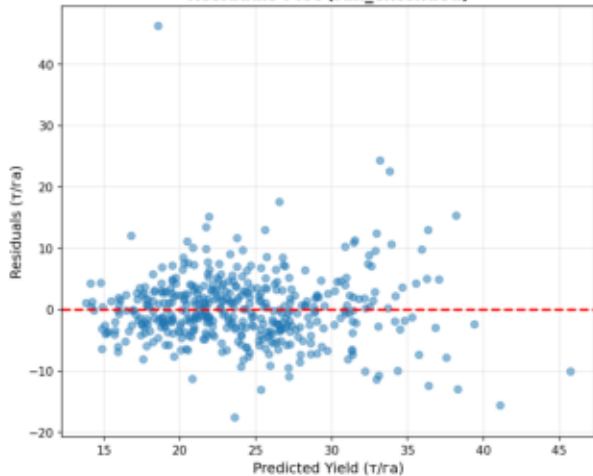
Feature



Predictions vs Actual (full_extended / all_no_leak_oof)



Residuals Plot (full_extended)



Dataset statistics

Dataset	N obs	N fields	Years	Mean yield (t/ha)	Std yield (t/ha)	N features
base_with_ndvi	440	30	2010–2025	24.04	7.69	37
ndvi_only_extended	441	30	2010–2025	24.08	7.74	51
full_extended	441	30	2010–2025	24.08	7.74	41
ops_ndvi_2021_2025	136	30	2021–2025	30.16	7.81	60